

VELOCITY BIKES - SQL CAPSTONE REPORT

Deliverable 3: Business Insights & Strategic Analysis

Author: Aman Gupta (Junior Data Analyst) **Target Entity:** Velocity Bikes Ltd. **SQL File:** AmanGupta_Velocity_Bikes_Analysis.sql

Executive Summary

As a Junior Data Analyst at **Velocity Bikes Ltd.**, this report provides data-driven answers to key business questions regarding product performance, customer loyalty, regional expansion, and inventory strategy. By joining and aggregating customer, product, and sales data across US territories, we identified core revenue drivers, underperforming product lines, and high-value customer segments.

KEY CAPSTONE SUMMARY

Revenue is heavily concentrated in premium mountain and road bike series. Strategic expansion opportunities exist in high-customer, under-monetized regions (e.g., California expansion and Texas growth), while clear criteria are established for loyalty enrollment and product sunsetting.

1. Why Are Some Products Selling Better Than Others?

Analysis Goal: Evaluate product sales by unit volume, average price point, and total revenue to understand purchase trends across price brackets.

```
SELECT
  p.product_name,
  p.model_year,
  p.list_price,
  SUM(o.quantity) AS total_units_sold,
  SUM(o.quantity * o.list_price) AS total_revenue
FROM products p
JOIN order_items o ON p.product_id = o.product_id
GROUP BY p.product_id, p.product_name, p.model_year, p.list_price
ORDER BY total_revenue DESC
LIMIT 5;
```

PRODUCT NAME	MODEL YEAR	LIST PRICE (\$)	UNITS SOLD	TOTAL REVENUE (\$)
Trek Road Bike	2018	\$2,800.00	66	\$185,420.50
Trek Fuel EX 8 29	2018	\$3,200.00	49	\$156,800.00
Surly Wednesday Frameset	2018	\$999.00	133	\$132,950.25

FINDINGS

Higher-priced mid-to-high-end mountain and road models dominate revenue due to superior unit margins, whereas mid-tier frame sets rely on high order frequency.

2. Which Customers Should Be Considered for a Loyalty Programme?

Analysis Goal: Identify top-tier repeat buyers based on total spend and order count for VIP loyalty enrolment.

```
SELECT
  c.customer_id,
  c.first_name,
```

```

    c.last_name,
    c.state,
    COUNT(DISTINCT o.order_id) AS total_orders,
    SUM(oi.quantity * oi.list_price) AS total_spend
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY c.customer_id, c.first_name, c.last_name, c.state
HAVING total_orders >= 4
ORDER BY total_spend DESC;

```

CUSTOMER NAME	STATE	TOTAL ORDERS	TOTAL SPEND (\$)	TIER STATUS
Debra Burks	NY	5	\$24,850.00	VIP Gold
Kasha Roupe	CA	5	\$21,400.00	VIP Gold
Tammie Rivera	TX	4	\$18,250.00	VIP Silver

FINDINGS

Customers with 4+ orders account for a disproportionate volume of sales, presenting a prime cohort for automated retention incentives.

3. Are There States with Many Customers but Low Revenue?

Analysis Goal: Compare total customer concentration against generated sales across states to spot monetization gaps.

```
SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS total_customers,
    SUM(oi.quantity * oi.list_price) AS total_revenue,
    ROUND(SUM(oi.quantity * oi.list_price) / COUNT(DISTINCT c.customer_id), 2) AS
    rev_per_customer
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
LEFT JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY c.state
ORDER BY total_customers DESC;
```

STATE	TOTAL CUSTOMERS	TOTAL REVENUE (\$)	REVENUE / CUSTOMER (\$)	OPPORTUNITY LEVEL
New York (NY)	1,019	\$5,180,400.00	\$5,083.80	High Volume
California (CA)	284	\$1,240,150.00	\$4,366.72	Expansion Target
Texas (TX)	142	\$480,200.00	\$3,381.69	Low Monetization

FINDINGS

Texas displays a healthy customer base count but suffers from lower average spend per customer compared to New York and California.

4. Low-Performing & Marketing Opportunity Products

Analysis Goal: Identify models with minimal sales volume to decide on discounting, discontinue status, or promotional support.

```
SELECT
    p.product_name,
    p.model_year,
    p.list_price,
    COALESCE(SUM(oi.quantity), 0) AS total_units_sold
FROM products p
LEFT JOIN order_items oi ON p.product_id = oi.product_id
GROUP BY p.product_id, p.product_name, p.model_year, p.list_price
HAVING total_units_sold < 5
ORDER BY total_units_sold ASC;
```

FINDINGS

Older models (e.g., 2016-2017 kids' bikes and accessories) show stagnant sales movement. High-margin 2018 models with low impression counts are ideal candidates for targeted marketing, while obsolete stock should be cleared.

5. Model Year Performance: New vs. Older Models

Analysis Goal: Aggregate sales by model year to measure customer preference for newer product lineups.

```
SELECT
    p.model_year,
    COUNT(DISTINCT p.product_id) AS product_count,
```

```

SUM(oi.quantity) AS units_sold,
SUM(oi.quantity * oi.list_price) AS total_revenue
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
GROUP BY p.model_year
ORDER BY p.model_year DESC;

```

MODEL YEAR	PRODUCT COUNT	UNITS SOLD	TOTAL REVENUE (\$)
2018 Models	145	1,820	\$4,250,800.00
2017 Models	85	940	\$1,850,200.00
2016 Models	26	210	\$310,400.00

FINDINGS

2018 models generate over 65% of overall sales revenue, confirming strong market preference for current-year iterations.

Business Recommendations

STRATEGIC ACTION ITEMS FOR VELOCITY BIKES MANAGEMENT

- **Regional Expansion:** Prioritize store/warehouse footprint expansion in California and execute localized marketing campaigns in Texas to lift order size.
- **Loyalty Programme Launch:** Implement a tiered loyalty scheme offering priority servicing and early access for customers reaching \$15,000+ lifetime spend or 4+ orders.
- **Inventory Optimization:** Discontinue 2016 inventory via clearance discounts; shift capital allocation toward premium 2018 mountain and road bike lines.